



2.4.7 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

My favorite mistake from today was ...

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Unit 2, Lesson 5: Adding and Subtracting Polynomials



Benchmark

MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.



Learning Target

- ☐ I can add and subtract polynomials.



2.5.1 Warm-Up: Math Fail

1. The classified advertisement shown is an example of a *math fail*.



What do you notice about the advertisement that makes it a math fail?



2.5.2 Guided Instruction: Adding Polynomials



Like terms are those whose variable(s) and their corresponding exponents are the same.

The following are examples of like terms.

Example 1	Example 2
$3.7mn^6$ and $-12mn^6$	$-cb^2a^{10}$, $81a^{10}cb^2$, and $\frac{1}{2}a^{10}b^2c$

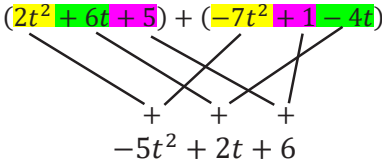
1. Discuss with your partner why $5ev^9$ and $5ev^3$ are not like terms. Summarize your discussion.
2. Draw an arrow to match the like terms between column A to column B. Then, check your matches with your partner.

Column A
$4xy^6$
$3xy$
$\frac{11}{4}x^5y^5$
$4x^6y$
$-1.6x^2y^3$
$-6\frac{7}{8}x^3y^2$

Column B
$9yx$
x^6y
$0.2x^3y^2$
$-2.9x^2y^3$
$-7y^6x$
$-3x^5y^5$

When adding and subtracting polynomials, only like terms can be combined.

3. Ziare and Colby used different strategies to add the polynomials $(2t^2 + 6t + 5)$ and $(-7t^2 + 1 - 4t)$.
- a. Work with your partner to describe what each student did and identify one helpful aspect used in their strategy.

$(2t^2 + 6t + 5) + (-7t^2 + 1 - 4t)$		
	Ziare's Strategy	Colby's Strategy
		$ \begin{array}{r} 2t^2 + 6t + 5 \\ + \quad -7t^2 - 4t + 1 \\ \hline -5t^2 + 2t + 6 \end{array} $
Description of the Strategy		
What was helpful?		

- b. Find the sum of $(19h^4 - 2h + 25k) + (8h - 14k + h^4)$ using both Ziare's and Colby's strategies.

$(19h^4 - 2h + 25k) + (8h - 14k + h^4)$	
Ziare's Strategy	
Colby's Strategy	

4. Find the sum using your preferred method. Write the sum in standard form.

$$(-3p^3k - 2k^2 + 1.4p) + (2.7p^3k - 4.6p + 5.8k^2)$$





2.5.3 Collaboration: Subtracting Polynomials

The process for subtracting polynomials is similar to the one for adding polynomials.

- With your partner, find the difference of $(8x^5 + 4x^3 - 7)$ and $(11 - x^3 + 5x^5)$ by completing Ziare's and Colby's strategies.

$(8x^5 + 4x^3 - 7) - (11 - x^3 + 5x^5)$	
Ziare's Strategy	Colby's Strategy
$(8x^5 + 4x^3 - 7) - (11 - x^3 + 5x^5)$	$ \begin{array}{r} 8x^5 \quad +4x^3 \quad -7 \\ -(5x^5) \quad -(-x^3) \quad -(+11) \\ \hline \end{array} $

- Oliver noticed another strategy that can work for subtraction.

He said, "I know the subtraction sign, or negative, can also mean 'opposite,' so I read the problem as $(8x^5 + 4x^3 - 7)$ plus the opposite of $(11 - x^3 + 5x^5)$."

He rewrote the problem to model his thinking.

$$(8x^5 + 4x^3 - 7) + (-11 + x^3 - 5x^5)$$

- Explain why you agree or disagree with Oliver's thinking.
- Why might it be helpful to rewrite subtraction of polynomials as addition of polynomials?

3. Find the difference of $(5m^2 - 4)$ and $(8m + 4 + 3m^2)$.
Write the resulting polynomial in standard form.



2.5.4 Guided Practice: Adding and Subtracting Polynomials

Find each sum or difference. Write the result in standard form.

$$(14m^2n^2 + 2mn - 9) - (5m^2n^2 - 6mn + 17)$$

$$(-2.3r^2 + 8.12pr - 0.4p) + (-7pr - 5.2r^2)$$



2.5.5 Your Turn!

Find each sum or difference. Write the result in standard form.

1. $(ab^2 + 13b - 4a) - (7b + a + \frac{1}{3}ab^2)$

2. $(8m^5 + 4m^3 - 9) - (11 + 5m^5 - m^3)$

3. $(-9d + 5d^2 + 3) + (5d - d^2 + 13)$

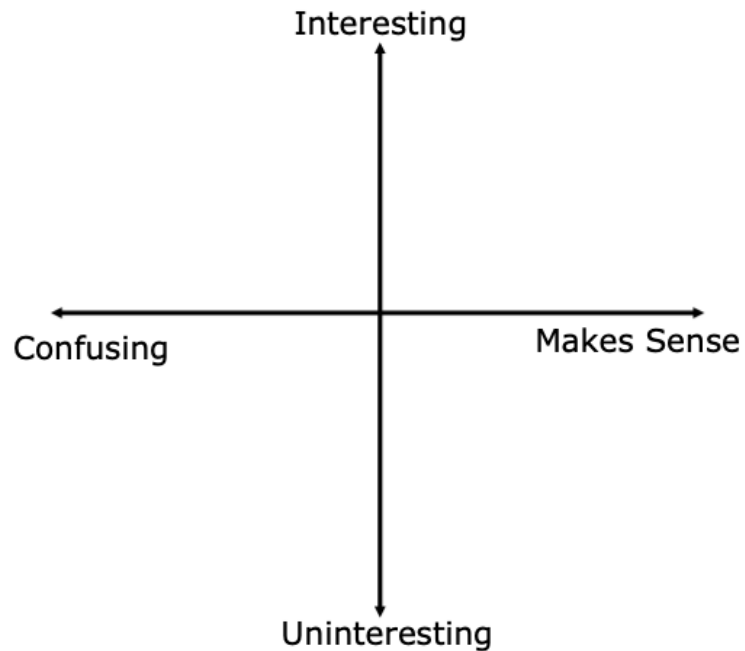
4. $3x^5y^3 + (x^3y^5 - 3x^5y^3)$

5. $(7.1ct - 5) + (3t + 4ct^2)$



2.5.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants to indicate how you feel about today's learning target, "I can add and subtract polynomials."



2. Describe why you plotted your point in that location.



Unit 2, Lesson 6: Multiplying Binomials With Algebra Tiles



Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.



Learning Target

- ☐ I can multiply binomials using algebra tiles.